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RESEARCH ARTICLE

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Thermal decomposition mechanism analysis of circulating solids sampled from Ende pulverized-coal gasifier

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Abstract

Ende pulverized-coal gasifier (EPCG) is a type of circulating fluidized bed gasifier, in which the circulating solids are resent into the furnace through a return conduit for regasification. Since the solid circulation in the return conduit mainly relies on gravity self-balancing, it is easy to cause low and unstable solid circulation rate and increase unburned combustible of fly ash. To solve these problems, a circulation enhancement technique using an ejector to provide a stable pressure barrier to promote solid circulation was proposed. To verify its feasibility, considering that the circulating solid-CO_2 reaction is a rate-controlling step in the gasification process, it is of great significance to explore the reactivity characteristics of the circulating solids under CO_2 atmosphere. In this work, thermal decomposition characteristics of Hailar circulating solids (HLE-CS), Hailar char (HLE-C), and Shenhua circulating solids (SH-CS) were investigated by non-isothermal thermogravimetric analysis to compare the gasification reaction characteristics of circulating solids and char. The HLE-CS and SH-CS used in the experiments were taken from the return conduit of 45,000 and 20,000 Nm^3/h EPCGs, respectively, while HLE-C was made from the raw coal of 45,000 Nm^3/h EPCG. The effects of different samples under the same thermal decomposition temperature range were investigated. The kinetic parameters were determined by the distributed activation energy model. The results indicated that remarkable peaks were found in the relationship between temperature (T) and activation energy (E) for HLE-CS and SH-CS, but there was no significant peak for HLE-C, and kinetic compensation effects also existed between $\ln(k_0)$ and E for different samples. The circulating solids of HLE-CS and SH-CS were more easily gasified than that of HLE-C in the lower temperature ambience. Moreover, HLE-CS and SH-CS have obvious pyrolysis and gasification stages, which can be completely introduced into EPCG for regasification, and increasing the circulating solid rate and temperature within a certain range is beneficial to enhance the gasification rate.

KEYWORDS

circulating solids, coal gasification, ejector, gasification rate, pulverized coal, thermal decomposition

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1 | INTRODUCTION

Circulating fluidized bed gasifier (CFBG) converts coal into fuel gas, like industrial fuel gas, city gas, and chemical syngas. Ende pulverized-coal gasifier (EPCG) is a typical CFBG,[1] with the advantage of being compatible with low-rank lignite.[2] Therefore, it is suitable for enterprises to produce syngas to meet the growing demand of fuel gas. In the EPCG, the raw coal is pyrolyzed and gasified in the furnace to produce circulating solids, which are returned to the reactor through the return pipe for regasification. Circulating solids are crucial for sustaining coal gasification within EPCG. However, the return conduit of EPCG is generally not equipped with a feedback control device (FCD). The flue gas and particle flow can easily flow back to the cyclone separator through the return conduit, thereby reducing the solid circulation rate, even causing an increase of unburned combustible of fly ash and fly ash discharge, and reducing the economic benefits of EPCG.

Concentrating on the study of solid circulating characteristics and its influence on the reaction process, Liu et al. and Bi et al. reported that CFBG operating and structural parameters had an influence on circulating solids rate, respectively.[3,4] Gül and Özdogan and Bhusarapu et al. discussed the effect of circulating solids rate on the stable operation and gasification process of CFBG system.[5,6] For the influence of circulating solids on the circulating fluidized bed (CFB), Zhang et al. gave a conclusion that the initial temperature of the circulating solids had an obvious influence on the devolatilization process.[7] To increase the carbon conversion, Pissot et al. returned the solids that came from the gas-phase outlet of the separator to the gasifier. The result revealed the potential of circulating solids as an activity enhancer in industrial dual fluidized bed gasification systems.[8] The above circulating solids came from the outlet of the separator instead of the solids in the return conduit of EPCG.[7,8] To solve the problems of low and unstable solid circulation rate of EPCG, Liu et al. proposed a circulation enhancement technique (Figure 1a), in which an ejector was employed as FCD to provide a stable pressure barrier to promote solid circulation.[9] For this technique, the gasification mechanism of circulating solids, as well as the influence of increased circulating solids on operation stability in the furnace, should be further studied.

In the process of pulverized coal gasification, the gasification mechanism of circulating solids under practical EPCG operation conditions need to be studied comprehensively. The gasification process generally includes volatile release and char gasification. The pyrolysis affects particle porosity and surface morphology, which is the basis of subsequent thermal decomposition process.[10] During the circulating solid gasification in a practical EPCG, the circulating solid-CO_2 gasification rate is much slower than steam and oxygen gasification rate; it is thus considered as the rate-controlling step in the gasification process. It is of great significance to understand the CO_2 gasification kinetic parameters to the design and optimization of EPCG.

Thermogravimetric analysis (TGA) is a technique widely used to study the characteristics of pyrolysis and

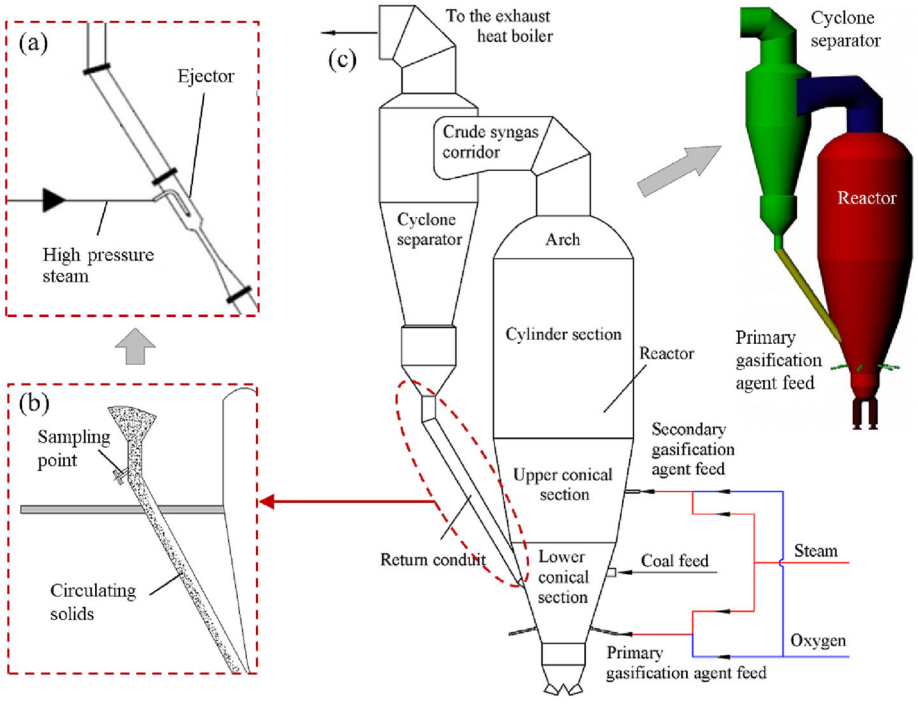


FIGURE 1 Schematic diagram of Ende pulverized-coal gasifier and the sampling point of circulating solids: (a) Circulation enhancing technology; (b) circulating solids sampling point; (c) schematic diagram of the EPCG.

gasification and to determine reaction kinetics.[11–13] Experimental data collected from TGA can be applied to a kinetic model to obtain the kinetic parameters of thermal decomposition processes. The kinetic compensation effects have been reported during the reactions, and various explanations have been proposed to explain the reaction mechanisms.[14–16] Non-isothermal methods are widely used in studying thermal decomposition kinetics due to their simplicity.[17,18] Meanwhile, non-isothermal methods can get rid of chemical and physical changes during the test. Thermal decomposition kinetic models include three pseudo-component model,[19] three-parallel Gaussian reaction model,[20] random pore,[21] and distributed activation energy model (DAEM).[22–24] The DAEM can effectively simulate the entire process of complex thermal decomposition and thus has been widely used for decomposition simulation of many solid fuels, that is, low-rank coal, biomass, and combustible solid waste. Miura and Maki further simplified the model to estimate the activation energy and corresponding pre-exponential factor for at least four TGA curves obtained at different heating rates.[25] For thermal decomposition of samples under a non-isothermal CO_2 atmosphere, Skodras et al. applied the first-order volume reaction model to determine the kinetic parameters of low-rank coal chars in isothermal coal-CO_2 gasification.[26] To study the gasification reaction characteristics, our previous work applied the DEAM model to analyze the kinetic parameters and investigated the reaction characteristics of coal char in isothermal char-CO_2 gasification, where the chars (C500 and C850) that simulated different temperatures in the gasifier were produced in the laboratory.[27] It can be noticed that few studies have investigated the circulating solids produced by industrial CFBG reactors. Therefore, it is necessary to make a further study on the thermal decomposition of circulating solids.

In this study, focusing on the technique proposal of using an ejector on the EPCG to increase the solid circulating rate, the circulating fly ash gasification mechanism is obtained and the feasibility of the technique is examined. The kinetics characteristics of circulating solids taken from the return conduit of 45,000 and 20,000 Nm^3/h EPCGs and char produced from coal taken from the 45,000 Nm^3/h EPCG are investigated under a CO_2 atmosphere. The DAEM is adopted to analyze the experimental TGA data. The main objectives of this study are to determine: (1) The pyrolysis and gasification characteristics of circulating solids; (2) the gasification mechanism and difficulty of increased circulating solids in EPCG, and the activity degree of gasification reaction between circulating solids and CO_2; (3) the influence of increased circulating solids on operation stability in furnace.

2 | METHODOLOGY

2.1 | Sample preparation

Figure 1 shows a schematic diagram of EPCG and the sampling point of circulating solids. EPCG is mainly composed of reactor, cyclone separator, and return conduit. Pulverized coal is fed into the bottom part of the reactor via a coal feeding channel. The primary and secondary gasification agents composed of oxygen and superheated steam enter the reactor through their respective gasification nozzles. In the experiments, Hailar circulating solids (HLE-CS) and Shenhua circulating solids (SH-CS) were taken from two 45,000 and 20,000 Nm^3/h EPCGs using Hailaer coal and Shenhua coal as raw coal, respectively. The temperature range in the furnace was 900–1000°C, and the average temperature was around 960°C. During each EPCG run, the sampling pipe (Figure 1b) located at the upper end of the return conduit was opened to sample the circulating solids. The sampling rate was around 1 kg/min, and the sampling weight was about 5 kg. Considering that the proximate analysis and ultimate analysis of Hailar coal and Shenhua coal were relatively close, and this study mainly focused on the thermal decomposition mechanism of circulating solids, only one kind of coal char was selected for comparative analysis. Hailar char (HLE-C) was produced from the raw coal of the 45,000 Nm^3/h EPCG. For HLE-C, a crucible containing 3 g of the coal sample (<125 μm) was placed in the cooled upper zone of the reactor before heating from ambient temperature to 500°C under a N_2 atmosphere (2 L/min). The crucible was then rapidly pushed into the constant temperature zone and held at the selected temperature for 30 min under a N_2 atmosphere. Then, the crucible was returned to the cooled zone for 20 min to obtain the char sample. The reason why HLE-C was produced at 500°C was that pyrolysis occurred at this temperature, and most of the volatiles were released, and good coal char can be obtained for subsequent TGA tests. Before the TGA tests, the HLE-CS and SH-CS samples were placed in a drying oven at 105°C for 4 h to remove external moisture. Samples were ground and sieved to less than 125 μm in diameter.

2.2 | Measurement of non-isothermal thermal decomposition reactivity under CO_2 atmosphere

Thermal decomposition of HLE-CS, HLE-C, and SH-CS under a CO_2 atmosphere was investigated with a STA449C thermal analyzer (Netzsch-Gerätebau GmbH, Germany) in thermogravimetric differential scanning

calorimetry (TG-DSC) mode. Each prepared sample (approximately 5 mg) was loaded into an aluminum crucible and heated under a CO_2 atmosphere. The experiment was divided into three stages. In the first stage, the temperature was programmed from room temperature to 110°C. In the second stage, the sample was dried and dehydrated at a constant temperature of 110°C for 10 min. In the third stage, the temperature was set between 110 and 1250°C. The carrier gas flow rate was 60 mL/min. All of the samples were analyzed by TGA at four heating rates of 15, 25, 35, and 45°C/min. The range of heating rates (<50°C/min)[28–30] and four TGA curves[31,32] were used to investigate the thermal decomposition mechanism.

The conversion rate (V/V^*) of a sample during non-isothermal gasification can be determined by

$$\frac{V}{V^*} = \frac{W_0 - W_t}{W_0 - W_{\text{ash}}} \tag{1}$$

where V is the mass loss at time t, V^* is the mass loss at the end of thermal decomposition, W_0 is the sample mass before thermal decomposition, W_t is the sample mass at any time t during thermal decomposition, and W_{ash} is the remaining ash mass after complete decomposition.

2.3 | Kinetic analysis

The activation energy (E) and pre-exponential factor (k_0) in the Arrhenius equation shows a linear relationship, and the relationship is shown in Equation (2):

$$\ln(k_0) = aE + b \tag{2}$$

where a is the slope on the plot between $\ln(k_0)$ and E, and b is the y-intercept when $E = 0$.

The DAEM model is based on the infinite parallel reaction hypothesis, which assumes that many first-order reactions simultaneously occur in the reaction system. These reactions are irreversible and independent. For the same V/V^*, the activation energy of all reactions has the same k_0. The activation energy values corresponding to all reactions are considered to have continuous distributions. The conversion equation is given by

$$1 - V/V^* = \int_0^\infty \exp\left(-\frac{k_0}{a}\int_0^T e^{-E/(RT)}dT\right) f(E)dE \tag{3}$$

where α is the heating rate, R is the gas constant, T is temperature, and k_0 is the pre-exponential factor corresponding to each activation energy value. $f(E)$ is the distribution of the activation energy curve, which indicates the activation energy of the irreversible reaction of different orders. The distribution function value plays an important role in determining the volatile release rate.

In this study, both E and k_0 were obtained from four experiments using different heating profiles. The calculation was performed with

$$\ln\left(\frac{\alpha}{T^2}\right) = \ln\left(\frac{k_0 R}{E}\right) + 0.6075 - \frac{E}{RT} \tag{4}$$

The procedures used to estimate E and k_0 were as follows:

1. The V/V^* curves at four heating rates were experimentally measured;
2. The $\ln(\alpha/T^2)$ and $-1/(RT)$ values were calculated for the same V/V^*;
3. The $\ln(\alpha/T^2)$ values of several conversion curves under the same mass loss rate level were calculated. That is, $\ln(\alpha/T^2)$ was plotted against $1/T$, and the activation energy E and k_0 were obtained from the slope and intercept of the straight line.

3 | RESULTS AND DISCUSSION

3.1 | Elemental characteristics of HLE-CS, HLE-C, and SH-CS

The results of the proximate and ultimate analyses of HLE-CS, HLE-C, and SH-CS are given in Table 1. The fixed carbon (FC) contents of the HLE-CS, HLE-C, and SH-CS samples are 33.73%, 76.8%, and 36.5%, and the ash contents are 54.04%, 20.74%, and 55.21%, respectively. The content of hydrogen and oxygen in HLE-CS and SH-CS is higher than that in HLE-C, and the content of FC is lower than that in HLE-C. Comparing the volatile content of each sample, it can be noticed that the volatile content of HLE-C was as low as 2.46%, indicating that most of the volatile matter was released. However, the volatile contents of HLE-CS and SH-CS were much higher, which was due to the short residence time of pulverized coal inside EPCGs, resulting in incomplete release of volatiles and insufficient gasification reaction.

3.2 | Sample mass loss characteristics

Figure 2a–c shows $1 - V/V^*$ as a function of T for HLE-CS, HLE-C, and SH-CS at heating rates of 15, 25, 35, and

TABLE 1 Proximate analysis and ultimate analysis of HLE-C, HLE-CS, and SH-CS.

	Proximate analysis (wt.%)			Ultimate analysis (wt.%)					
	A	VM	FC	C	H	N	S[a]	O[b]	HHV, (kJ/kg)
HLE-CS	54.04	12.23	33.73	82.89	2.34	1.65	1.83	11.29	14366.35
HLE-C	20.74	2.46	76.80	97.83	0.59	1.11	0.23	0.24	23742.99
SH-CS	55.21	8.29	36.50	91.16	1.20	1.18	1.75	4.71	13621.24

Note: Proximate analysis and HHV were tested on the air dry base, and ultimate analysis was tested on the dry ash-free base.
[a]According to the GB/T 214-2007, obtained by Ehrlich card method.
[b]Obtained by difference.

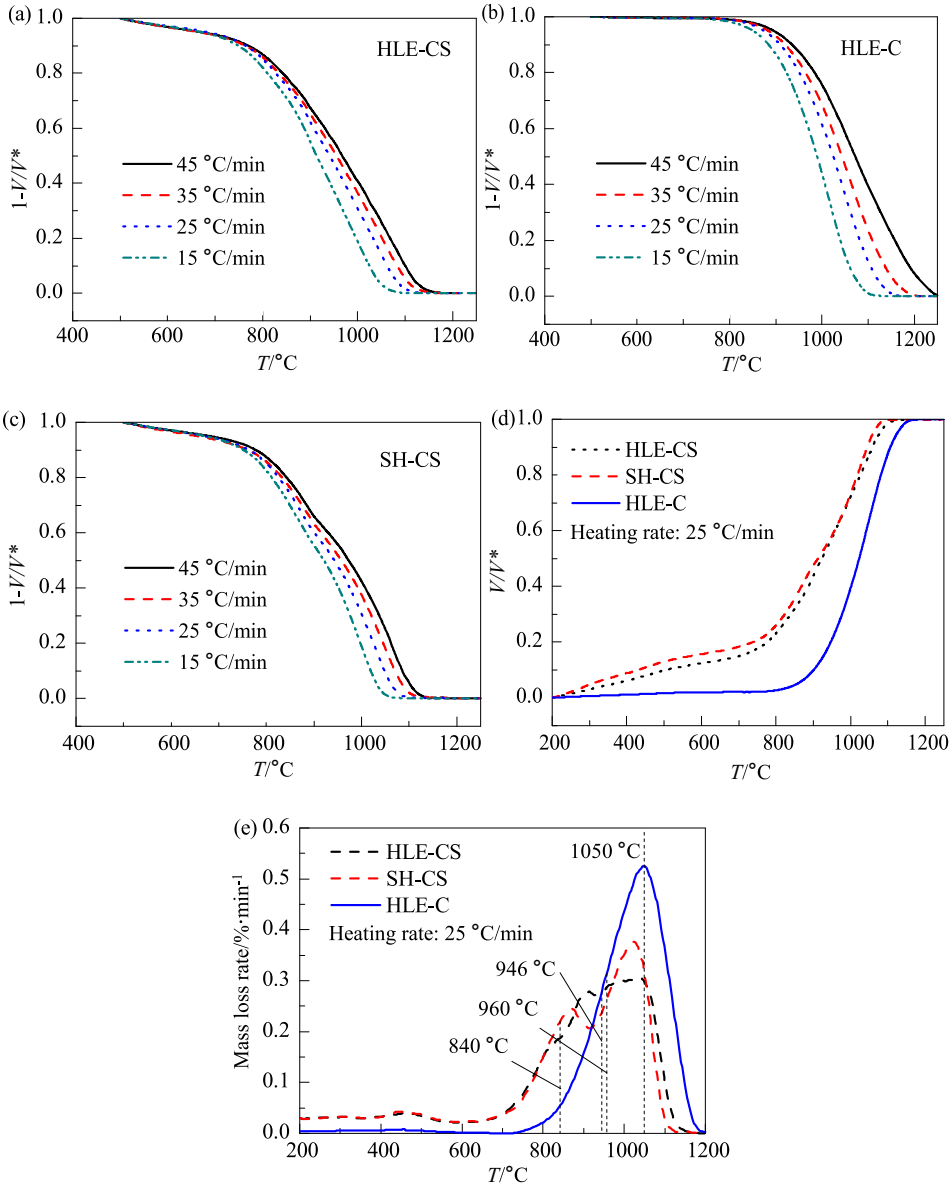


FIGURE 2 $1 - V/V^*$ as a function of T: (a) HLE-CS at heating rates of 15, 25, 35, and 45°C/min; (b) HLE-C at heating rates of 15, 25, 35, and 45°C/min; (c) SH-CS at heating rates of 15, 25, 35, and 45°C/min; (d) V/V^* at a heating rate 25°C/min; (e) their mass loss rates at 25°C/min.

45°C/min, respectively. V/V^* and the mass loss rate (d(V/V^*)/dt) at a heating rate of 25°C/min are shown in Figure 2d,e, respectively. As shown in Figure 2a–c, when the conversion ($1 - V/V^*$) is a fixed value, the thermogravimetric temperature increases with increasing heating rate. During the test, at a heating rate of 25°C/min, when the thermogravimetric mass loss of the sample was 2%–95%, the main decomposition temperature intervals of HLE-CS, HLE-C, and SH-CS were 500–1180, 660–1200, and 490–1150°C, respectively. The initial and final

thermal decomposition temperatures of HLE-CS and SH-CS were lower than those of HLE-C.

As shown in Figure 2d, at the same T for a heating rate of 25°C/min, when the temperature is lower than 1180°C, the V/V^* values of HLE-CS and SH-CS are similar, but higher than that of HLE-C. As shown in Figure 1e, the mass loss rate curves of HLE-CS and SH-CS show a similar trend with two obvious peaks of mass loss rate in DTG curves, whereas the curve of HLE-C contains only one prominent peak. The reason for this phenomenon was that due to the high volatile content in HLE-CS and SH-CS (8.29% and 12.23%, respectively), there was secondary pyrolysis of circulating solids during the decomposition process, so that two mass peaks of pyrolysis and gasification appeared. As for HLE-C, most of the volatiles were released during the production of HLE-C (Table 1), so there was no obvious volatile release process, and only one peak appeared. The peak (1025°C) was the maximum mass loss peak during fast decomposition when many fierce gasification reactions occurred. The main mass loss peak values of HLE-CS, HLE-C, and SH-CS occurred at 1030, 1050, and 1025°C with values of 0.312%·min^{-1}, 0.525%·min^{-1}, and 0.375%·min^{-1}, respectively. This shows that the maximum decomposition rate of HLE-C is higher than those of HLE-CS and SH-CS in the high-temperature zone.

Since the temperature in EPCG is around 960°C, the mass loss rate of circulating solids at the temperature of 960°C is also induced in Figure 2e. When the temperature was lower than 840°C, HLE-C had a rather slow char-CO_2 reaction rate; thus, the mass loss rates of HLE-CS and SH-CS were higher than that of HLE-C. In the temperature range of 840 to 960°C, the mass loss rate of HLE-C started to increase rapidly, and at the temperature of 946°C, the mass loss rate of HLE-C was higher than those of HLE-CS and SH-CS. In the temperature range of 960 to 1050°C, the mass loss rate of HLE-CS and HLE-C showed a significant increase, which indicates that the carbon conversion rate can be enhanced by increasing the operation temperature of EPCG in a certain temperature range (960–1050°C). The above results reflect that HLE-CS and SH-CS can be gasified at an average temperature of around 960°C in EPCG.

3.3 | Arrhenius curve analysis

Arrhenius plots of $\ln(\alpha/T^2)$ against $-1/RT$ at V/V* values of 0.05–0.95 are given in Figure 3a–c. According to the DAEM model, the E values can be obtained from four heating rate measurements. According to Equation (4), the E and k_0 values are determined from the corresponding Arrhenius plot.

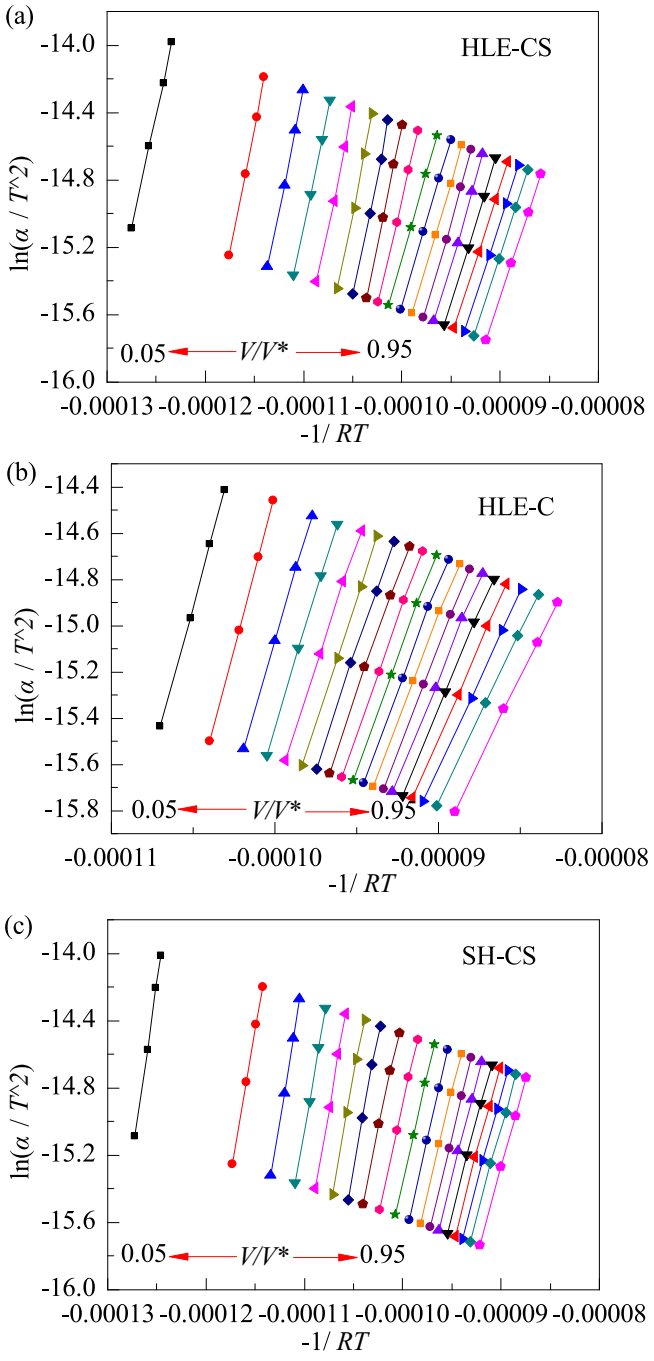


FIGURE 3 Arrhenius plots of $\ln(\alpha/T^2)$ against $-1/RT$ at V/V^* values of 0.05–0.95: (a) HLE-CS; (b) HLE-C; (c) SH-CS.

Based on the results and analysis in Figures 2 and 3, the stages of pyrolysis and gasification for each sample are determined and presented in Figure 4. The E and R^2 values of each sample are shown in Figure 4a–c, respectively. For HLE-CS and SH-CS, as shown in Figure 4a,c, the lowest value of R^2 is 0.88 when E is maximum. For HLE-C, as shown in Figure 4b, the trend is stable and the value of R^2 is above 0.98. For HLE-CS and SH-CS, the plots of E against T show a single-peak distribution. During the test, at the early stage of the reaction

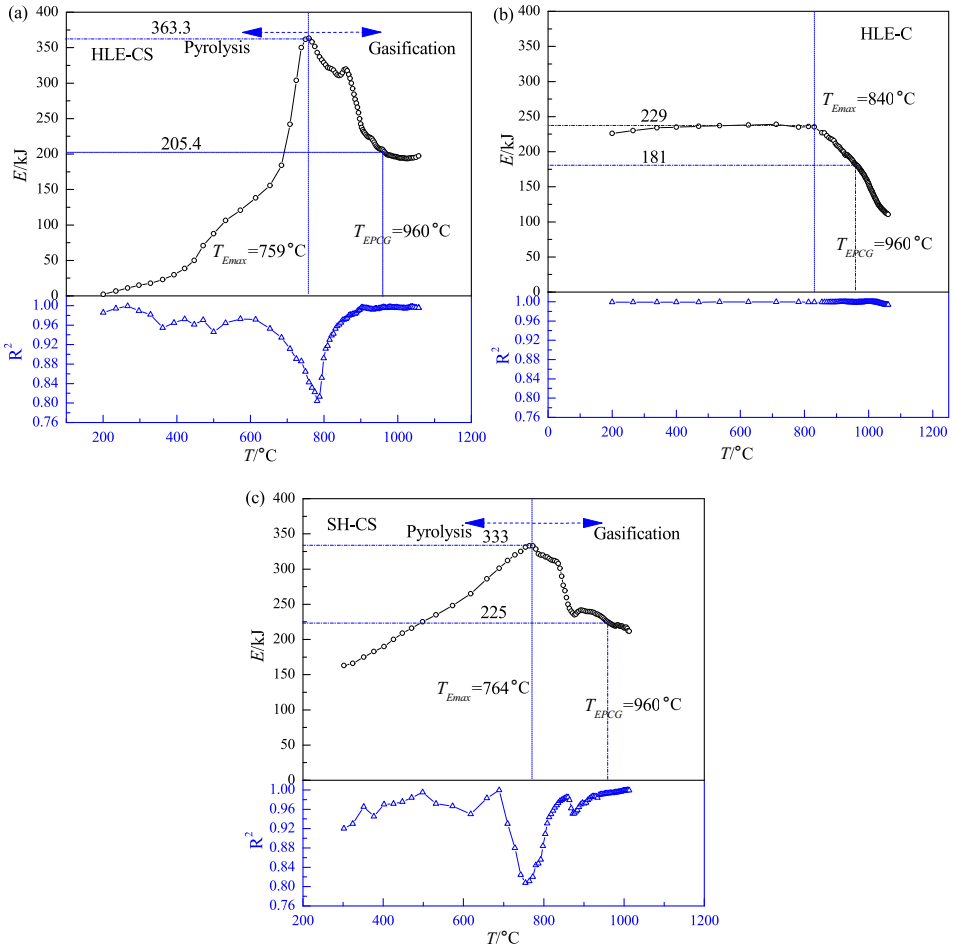


FIGURE 4 Relationships between E and T: (a) HLE-CS; (b) HLE-C; (c) SH-CS.

(HLE-CS < 759°C, SH-CS < 764°C), the E value increased with the increase of T. This is because although pyrolysis and gasification proceeded simultaneously in the early stage, the volatile release played a dominant role. At the same time, the carbon structure progressively condensed with the increase of T.[15] The bonds among condensed carbon structure required higher energy to break up, resulting in an increased E in the low temperature range. At the late stage of the reaction (HLE-CS > 759°C, SH-CS > 764°C), E decreased with the increase of T. That is because more-developed pore structure gradually formed as the temperature increased, resulting in an increase in its activated sites.[15] In this case, more CO_2 can enter the activation point by interdiffusion, which can increase the reaction rate.

For HLE-C, there is no significant peak during thermal decomposition (Figure 4b). As the temperature increases, the E value remains stable and then rapidly decreases, which is similar to the results in Wang et al.[27] That is because the volatile content of HLE-C was about 2.46%. From the flat curve in Figure 4b, it can be seen that due to the lower temperature in the early stage, the rates of volatiles release and char-CO_2 reaction are slow. When T > 840°C, the char-CO_2 reaction rate gradually increased, leading to the decrease of E value. At a temperature of 840°C, it can be seen that HLE-C lacked an obvious pyrolysis process, which was due to the low volatile content of HLE-C, and no obvious volatile release process appeared. There may be coexistence of pyrolysis and gasification, both of which showed certain compensation effects, but with a lower degree of pyrolysis.[27,33]

The E values at the temperature of 960°C are also shown in Figure 4a–c. The E values of HLC-CS, HLE-C, and SH-A are 205, 181, and 225 kJ, respectively. Compared with the E_{max} of HLC-CS, HLC-C and SH-CS, it decreased by 43.5%, 21.0%, and 32.4% at the temperature of 960°C. With the temperature increasing from 960 to 1050°C, the E values of HLE-CS, HLE-C, and SH-CS reduced to different degrees, and the decline rate of HLE-C was faster than those of HLE-CS and SH-CS.

3.4 | Pyrolysis and gasification characteristics

The value of $\ln(k_0)$ can be calculated by Equation (4). The relationships between E and $\ln(k_0)$ for HLE-CS, HLE-C, and SH-CS are shown in Figure 5a–c,

respectively. As shown in Figure 5a,c, for HLE-CS and SH-CS, the relationship between $\ln(k_0)$ and E shows good linear characteristic, exhibiting the so-called "kinetic compensation effect" during the pyrolysis and gasification stages.[26] As shown in Figure 5b, for HLE-C, there is a clear linear relationship in the dominant stage of gasification due to its low volatile content. The above results reveal that HLE-CS and SH-CS had obvious pyrolysis and gasification stages, which can be completely introduced into EPCG for regasification.

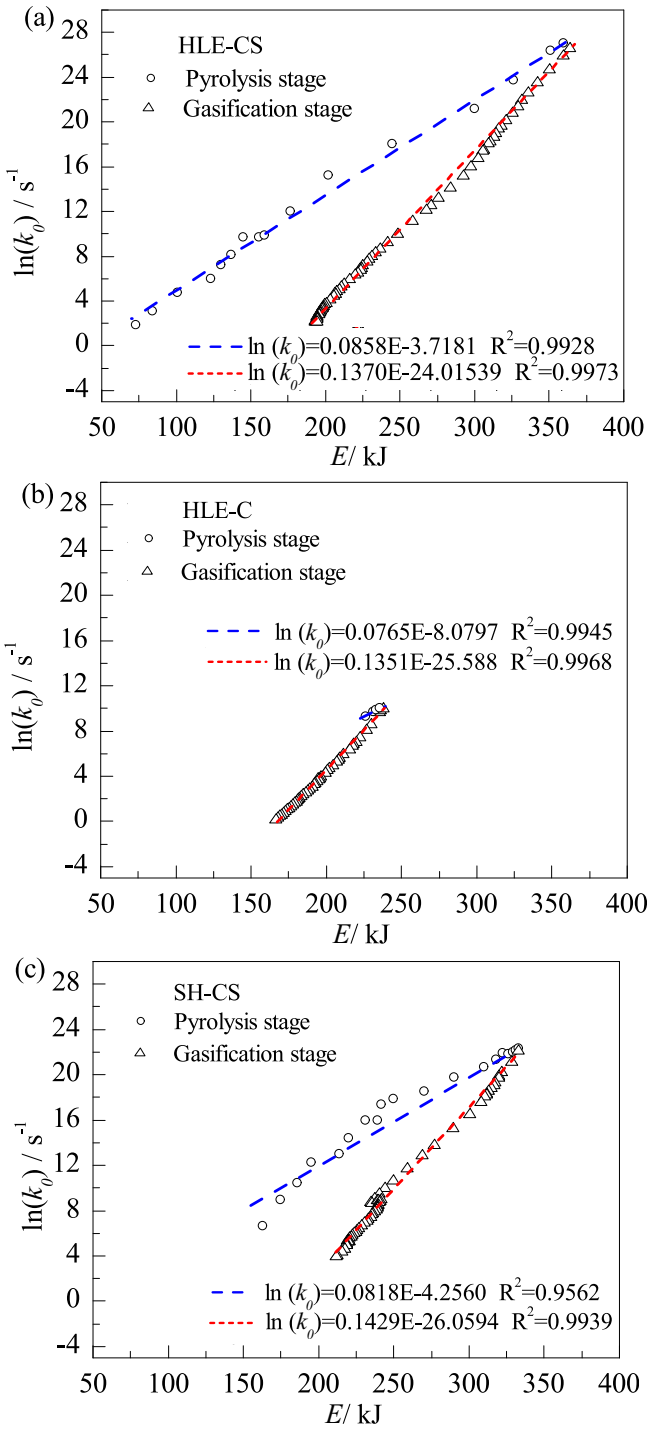


FIGURE 5 Relationships between E and $\ln(k_0)$: (a) HLE-CS; (b) HLE-C; (c) SH-CS.

Table 2 shows the values of a and b (Equation 2) of different samples. For more systematic comparison, test results from references are also given.[26,27] Under the same CO_2 atmosphere, the values of a were very similar to each other. For the value of b, its absolute value decreased along the lignite, bituminous coal, and low rank coal successively. This rule can provide a certain reference for the thermal decomposition mechanism analysis of different coal types. It can also be seen from Figure 5, in the gasification stage, the kinetic compensation effects of HLE-C produced from laboratory were almost the same as that of HLE-CS and SH-CS, indicating that the three samples under different conditions had the same reaction mechanism during the circulating solid (char)-CO_2 reaction.

For further comparison of the circulating solids (char)-CO_2 reaction mechanism of HLE-CS, HLE-C and SH-CS, the data of $\ln(k_0)$ and E are plotted in Figure 6. In the main gasification stage, with the temperature rising to 960°C, the E value of HLE-CS decreased from 363.7 to 205.0 kJ, and the value of $\ln(k_0)$ decreased from 22.0 to 6.1. For HLE-C, the E value decreased from 235.3 to 181.0 kJ, the value of $\ln(k_0)$ decreased from 9.6 to 1.9. For SH-CS, the E value decreased from 332.9 to 225.0 kJ, the value of $\ln(k_0)$ decreased from 26.5 to 4.4. When the temperature was lower than 840°C, HLE-CS and SH-CS exhibited higher E and $\ln(k_0)$. For the kinetic compensation effect, E indicates the average energy level of the carbon structure, and $\ln(k_0)$ indicates the number of active sites. Although the aromatic-ring cluster of HLE-CS and SH-CS with high condensation possessed a higher E, the number of active sites was also large. It leaded to the activation of many carbon atoms on the condensed aromatic ring cluster,[15] which resulted in the decrement of E and $\ln(k_0)$. However, when the temperature was lower than 840°C, the rates of volatile release and char-CO_2 reaction of HLE-C were slow (as shown in Figure 4). With the temperature rising from 840 to 960°C, the E and $\ln(k_0)$ values of the three samples decreased simultaneously. As the temperature further increased to 1050°C, the E and $\ln(k_0)$ values of the three samples decreased to some extent. The results are consistent with Figure 2d. This indicates that HLE-CS and SH-CS were more easily gasified than HLE-C in the lower temperature range, and can be further gasified in higher temperature (1050°C). The results reveal that at the operating temperature of EPCG, the two kinds of circulating solids were easier to react with CO_2 than raw coal char, and increasing the circulating solids rate and operation temperature in a reasonable range can enhance the gasification rate. In addition, due

TABLE 2 The factors of kinetic compensation effects for different samples.

	Skodras[26]	Wang[27]	SH-CS	HLE-C	HLE-CS
Atmosphere	CO_2	CO_2	CO_2	CO_2	CO_2
Samples	Low rank coal	Bituminous coal	Coal (from lignite)	Ash (from lignite)	Coal (from lignite)
a	0.1459	0.1335	0.1429	0.1351	0.137
b	−1.012	−13.28	−26.06	−25.59	−24.02
R^2	0.9987	0.99	0.9939	0.9968	0.9973

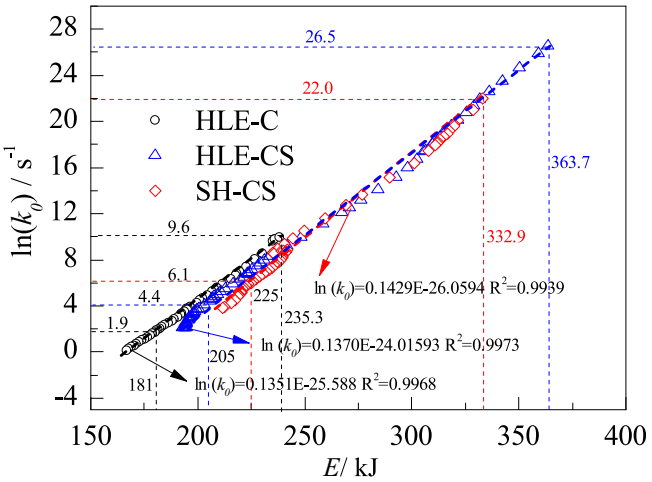


FIGURE 6 Relationships between *E* and ln(k_0): (a) HLE-CS in the gasification stage; (b) HLE-C in the gasification stage; (c) SH-CS in the gasification stage.

to the combustion heat release of volatiles in the circulating solids, the influence on the temperature distribution and operation stability in the furnace can be reduced.

4 | CONCLUSIONS

In this work, to verify the feasibility of the EPCG circulation enhancement technique, the thermal decomposition mechanisms of HLE-C, HLE-CS, and SH-CS under a CO_2 atmosphere were investigated by TGA. The conclusions are as follows:

1. When the temperature is lower than 840°C, the mass loss rate of HLE-CS and SH-CS is higher than that of HLE-C. When the temperature gets to 946°C, the mass loss rate of HLE-C is higher than those of HLE-CS and SH-CS. With the temperature further rising to 1050°C, the mass loss rate of HLE-CS and HLE-C shows a dramatic increase.
2. The circulating solids of HLE-CS and SH-CS have obvious pyrolysis and gasification stages, which can be completely introduced into EPCG for regasification. The circulating solids of HLE-CS and SH-CS are more easily gasified than that of HLE-C in the lower temperature range.
3. Increasing the circulating solid rate and temperature in a certain range is beneficial to enhance the gasification rate. Due to the combustion heat release of volatiles in the circulating solids, it can be concluded that the increased circulating solids have little impact on the temperature distribution and operation stability in EPCG.
4. The research results prove the feasibility of increasing the solid circulating rate to promote the gasification of pulverized coal. For the circulation enhancement technique that uses an ejector to promote solid circulation, in practical applications, the maximum amount of circulating solids should be obtained with the least amount of central injection gas to achieve low cost and stable operation of this technique.

Future work may be directed to investigate the effects of carrier gas velocity, gas composition, and temperature on the pyrolysis and gasification characteristics of circulating solids.

NOMENCLATURE

a	slope on the plot between ln(k_0) and *E*
b	y-intercept when the $E = 0$
E	activation energy, kJ/mol
k_0	pre-exponential factor, 1/s
R	gas constant, kJ/(mol·K)
T	temperature, °C
V	mass loss at time *t*, g
*V**	mass loss at the end of thermal decomposition, g
W_0	sample mass before thermal decomposition, g
W_{ash}	ash mass remaining after complete decomposition, g
W_t	sample mass at any time *t* during thermal decomposition, g
α	heating rate, °C/min

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